CAP6938 · SCIENTIFIC TRACK

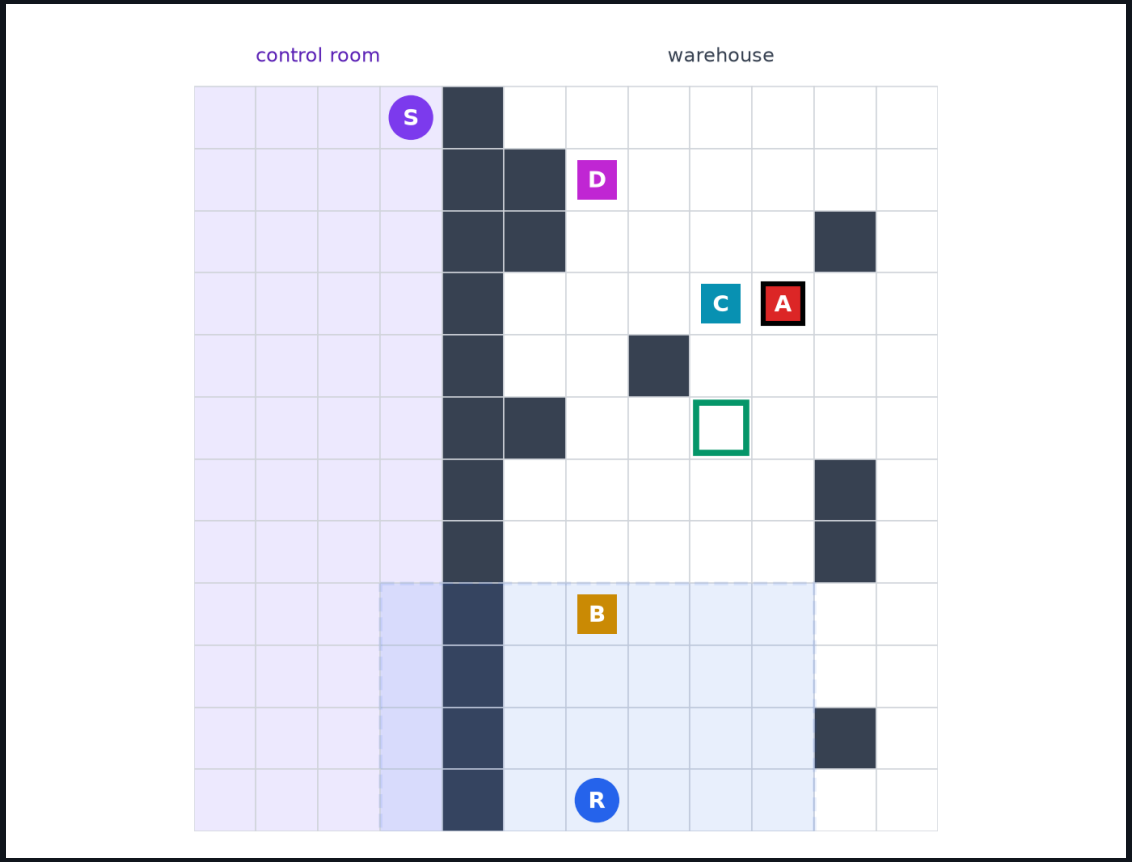
Can agents invent useful messages?

- Emergent Communication in a Grid World
- Christian Tadros · University of Central Florida
- Learned coordination, causal tests, and limits of transfer

INFORMATION STRUCTURE

The scout knows. The runner acts.

- A wall blocks movement between rooms
- Only the scout sees the target
- The runner sees nearby items and must deliver the right one



One targeted change to independent PPO

- Baseline: two separate recurrent PPO agents
- Modification: reward message information about the private cue
- No programmed codebook or listener supervision

PPO loss - lambda x cue information

Task-specific adaptation of positive signalling (Eccles et al., 2019)

ORIGINAL-METHOD CONTROL · POST-HOC EXPLORATORY

Does the added objective help?

- Unbiased IPPO: 22.7% held-out success
- Cue-directed IPPO: 87.6%
- Difference: +64.9 pp · 95% CI +63.3 to +66.4 pp

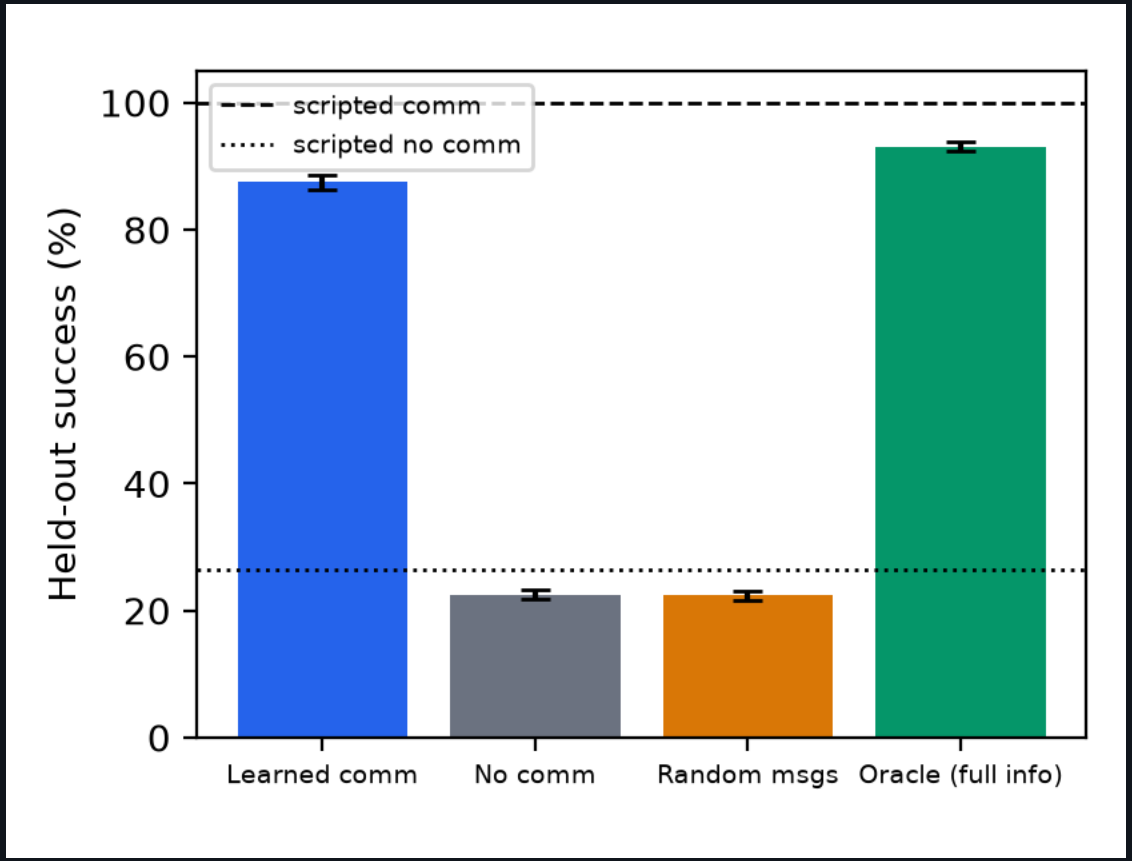
EVALUATION PROTOCOL

A benchmark of trained policies

- 20 independent training seeds per condition
- 200 fixed evaluation episodes per seed and split
- Final checkpoints only · 95% seed-bootstrap intervals

Communication improves held-out success

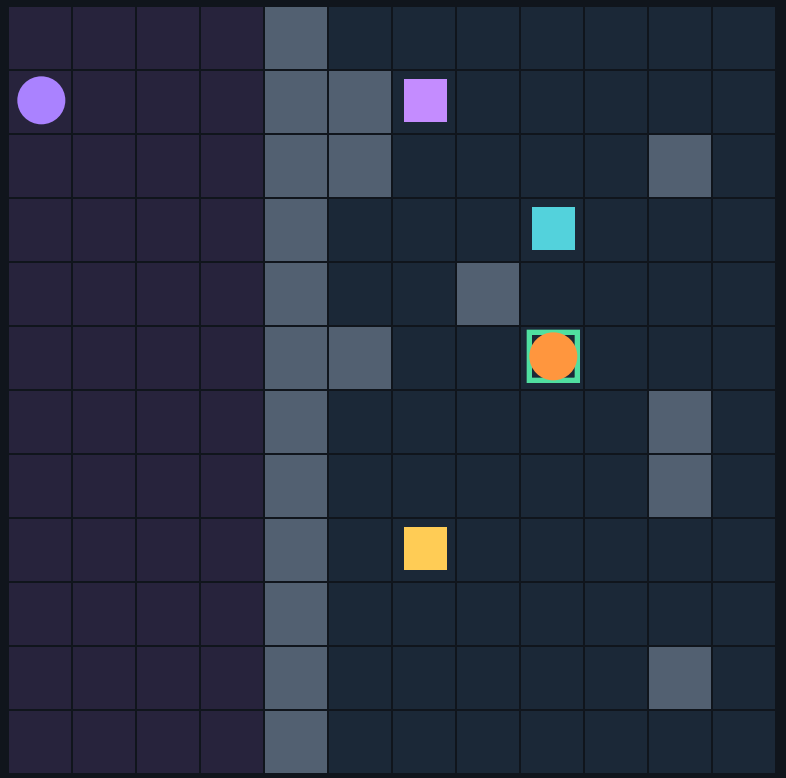
- Learned communication: 87.6%
- No communication: 22.5%
- Gain: 65.0 percentage points



SAME EPISODE · DIFFERENT RECEIVED MESSAGES

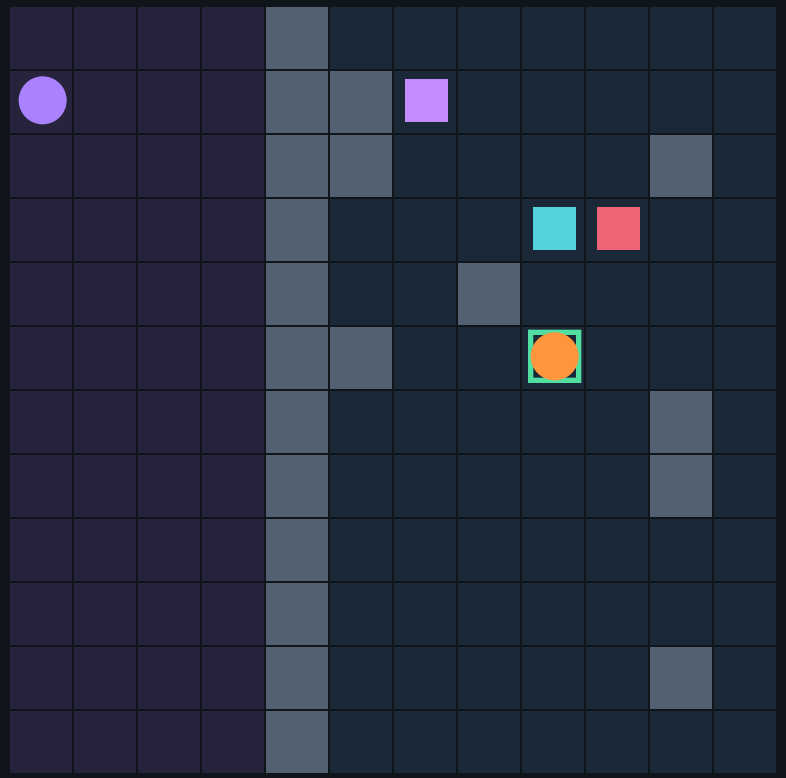
Does the listener use the meanings?

INTACT



CORRECT DELIVERY

PERMUTED

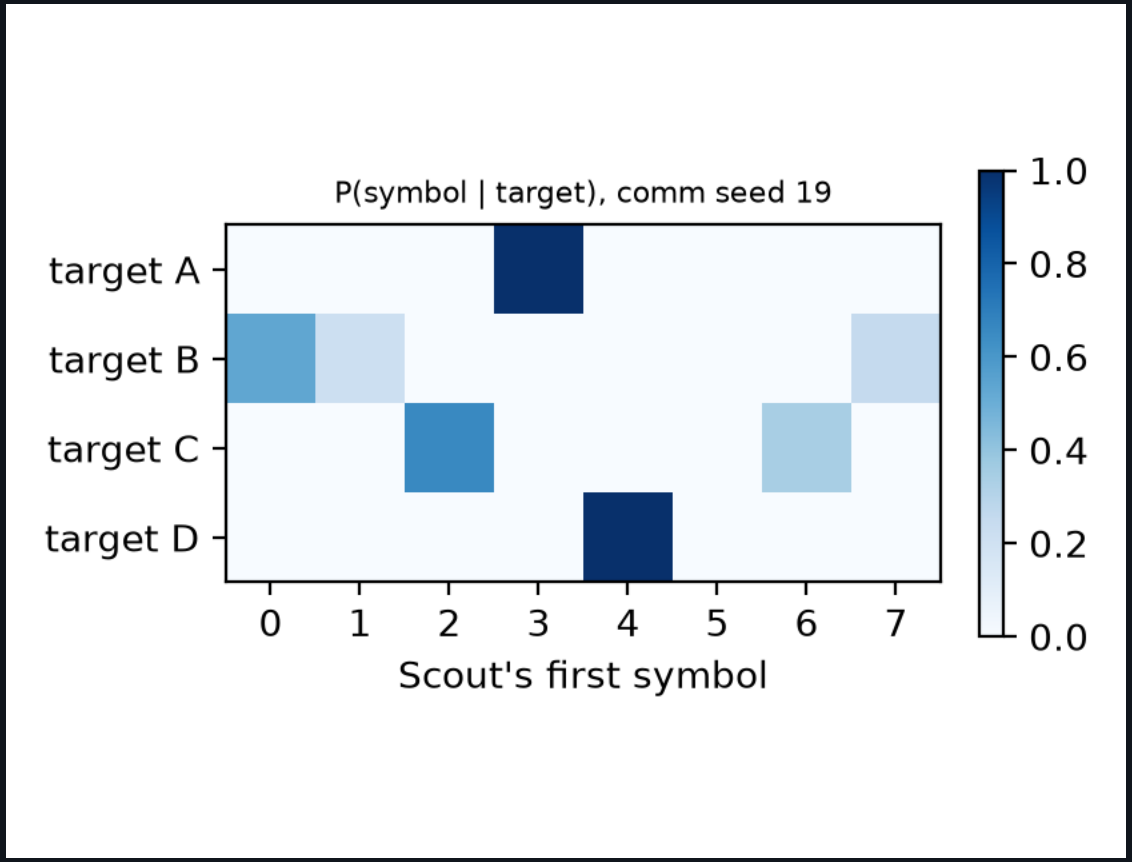


FAILURE

Held-out episode 0, representative seed 19. Terminal states; animate in record.html.

Informative speech is not enough

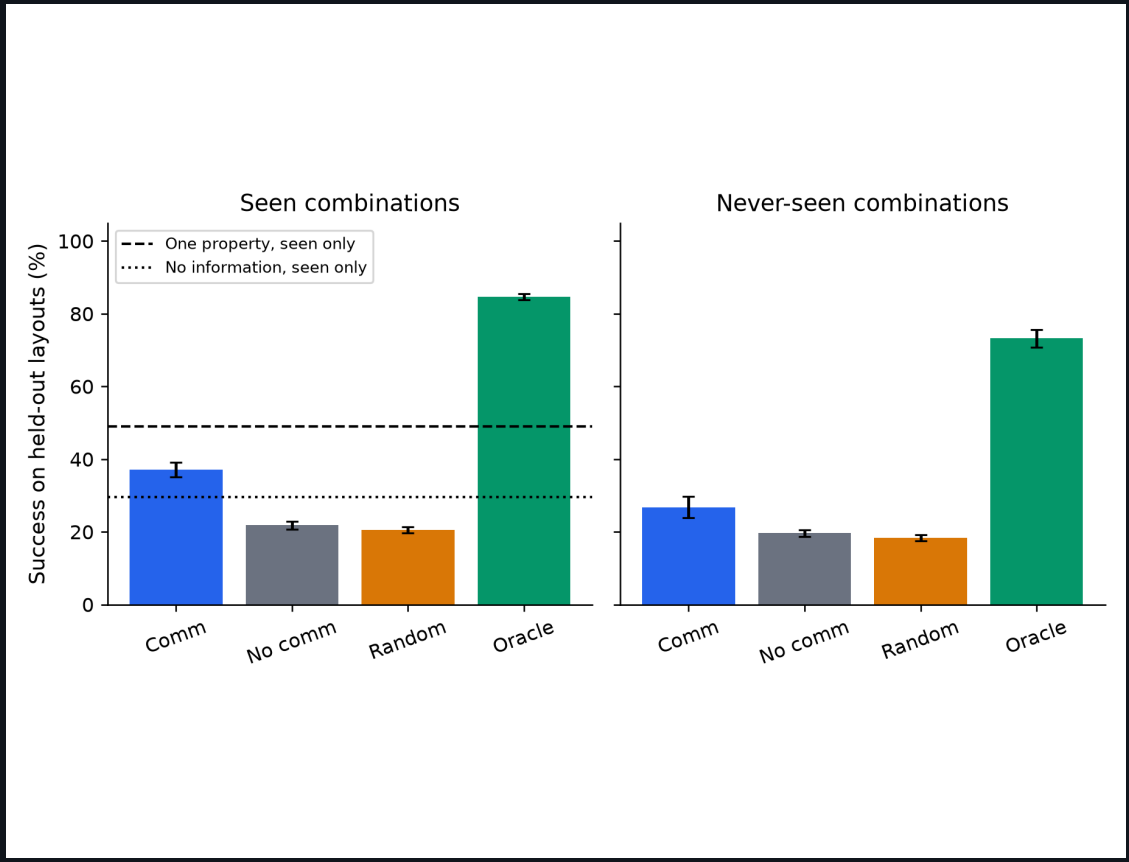
- Unheard scouts can still encode the target
- MI describes the speaker
- Interventions test whether the listener depends on the channel



V2 · SEPARATELY PRE-REGISTERED

Two words do not guarantee generalization

- Seen combinations: 37.1%
- Unseen combinations: 26.7%
- Communication supported · Generalization criterion not met



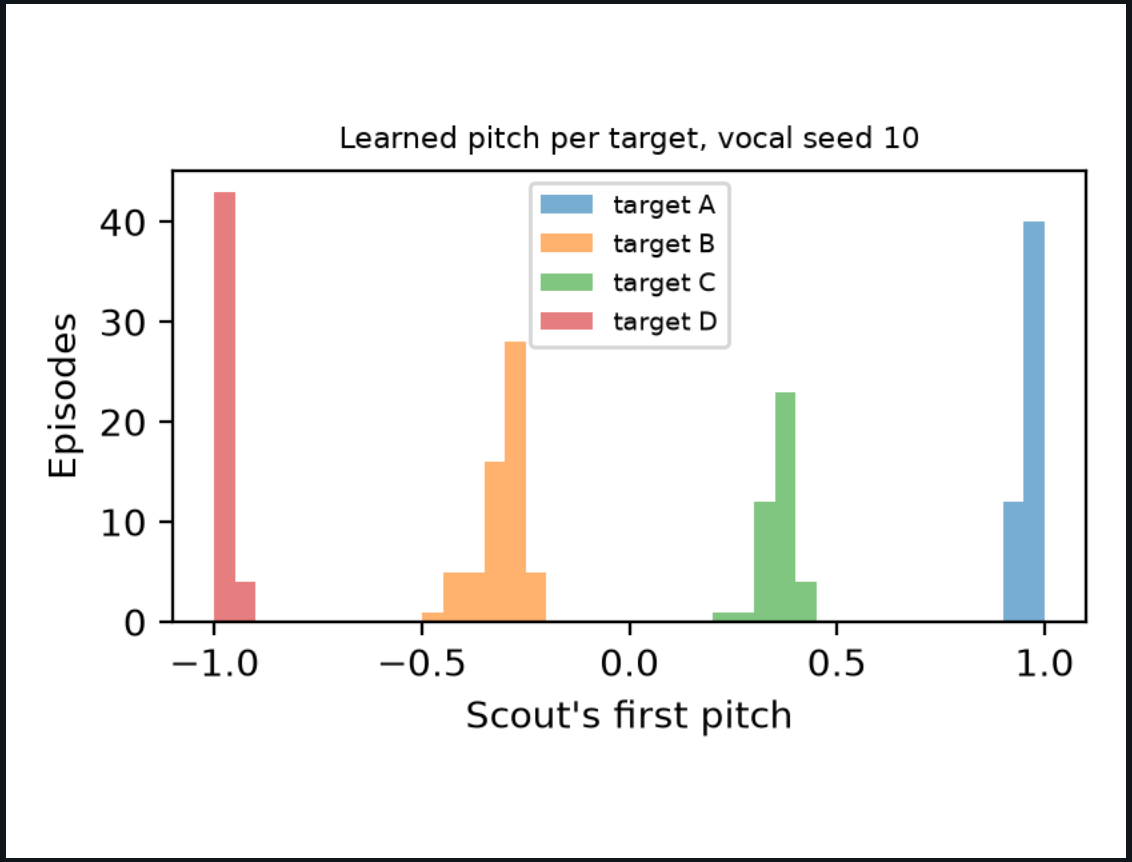
A LIMITATION IN THE TARGET PRIOR

Read the unseen result carefully

- The held-out combinations form a one-to-one matching
- Support-aware identification: 87.5% before navigation
- The fixed 50% reference does not prove compositionality

Exploratory: a continuous vocal channel

- Vocal communication: 73.0%
- Intact minus shuffled voice: 57.9 percentage points
- Distinct notes form · Cross-arm rates are descriptive



The implementation runs from saved weights

- Run RUN_DEMO.command, or the command below
- uv run python scripts/demo_inference.py --phase both
- New episode logs, metrics, receipt and offline replays

Useful coordination, with bounded claims

- Cue-directed signalling supports coordination in this task
- Causal message use differs from message informativeness
- The harder compositional generalization claim fails